Quiz: Dynamo

Test your knowledge of Dynamo by completing this quiz.

Quiz ?

Statement

In CAP theorem terms, Dynamo is ___ system?

Select Multiple Options

- [ ] AP (Available and Partition Tolerant)
- [ ] CP (Consistent and Partition Tolerant)
- [ ] CA (Consistent and Available)

Check Answer

Quiz ?

Statement

Amazon Dynamo is called a zero-hop DHT because:

Select Multiple Options

- [ ] A collision-free hash function is used.
- [ ] Consistent hashing allows the system to grow.

- [ ] Clients know the complete set of servers and can directly contact the server that holds the data they need.
- [ ] Each server node maintains a finger table that identifies other nodes.

Check Answer

Quiz ?

Statement

Does Dynamo always write data to the first 'N' nodes in the preference list?

Select Option

- ○ True
- ○ False

Check Answer

Quiz ?

Statement

How does Dynamo resolve conflicts between different versions of data?

Select Multiple Options

- [ ] Merkle tree
- [ ] Hinted handoff
- [ ] Vector clock
- [ ] Gossip protocol

Check Answer

Quiz ?

Statement

What strategies can be used to resolve data conflicts? Select all that apply.

Select Multiple Options

- [ ] Last write wins
- [ ] Conflict-free replicated data types (CRDTs).
- [ ] Sloppy quorum
- [ ] Vector clocks

Check Answer

Quiz ?

Statement

How does Dynamo handle temporary node failures? Select all that apply.

Select Multiple Options

- [ ] Merkle tree
- [ ] Hinted handoff
- [ ] Sloppy quorum
- [ ] Gossip protocol

Check Answer

Quiz ?

Statement

What are the benefits of Vnodes in Dynamo? Select all that apply.

Select Multiple Options

- [ ] Vnodes help spread the load more evenly across the physical nodes on the cluster. This speeds up the rebalancing process after adding or removing nodes.
- [ ] Vnodes enable the use of virtual machines to help Dynamo scale better.
- [ ] Vnodes make it easier to maintain a cluster containing heterogeneous machines.
- [ ] Use of Vnodes decreases the probability of hotspots, as Vnodes help assign smaller ranges to each physical node.

Check Answer

Quiz ?

Statement

What is optimistic replication in Dynamo?

Select Multiple Options

- [ ] When replicating, Dynamo chooses the closest nodes to the coordinator node to store the replicas.
- [ ] Dynamo tries to distribute replicas on different racks.
- [ ] Replicas may diverge and are not guaranteed to be identical at all times.
- [ ] All replicas are updated atomically, ensuring ACID semantics.

Check Answer

Quiz ?

Statement

Dynamo uses Vector clocks to:

Select Multiple Options

- [ ] Impose a total ordering of all get and put operations in the system.
- [ ] Disallow concurrent updates of the same object by choosing the earliest writer.
- [ ] Determine the sequence of write operations across multiple objects.
- [ ] Identify conflicting versions of the same object stored on different servers.

Check Answer

Quiz ?

Statement

Dynamo optimizes key lookups via:

Select Multiple Options

- [ ] Consistent hashing
- [ ] Finger tables
- [ ] Storing the table of all nodes in the system at each node.
- [ ] Storing all key-value sets in a hash table at each node.